Practice workbook · Match hormones to glands and effects, and contrast them with nerves

Naming the gland or system producing insulin, glucagon, ADH, adrenaline or thyroxine, stating their effects, and comparing hormonal with nervous signalling.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATCH-HORMONE-SOURCE-AND-EFFECT

1. Question 1 · part whole · 7 marks

November 2024 Paper 1

Worked-example retry · Attempt part whole only.

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1 The table gives some body processes.
Each substance or structure listed in the box is responsible for a body process.

Use information from the box to complete the table.

ADH	adrenaline	cerebellum	cerebrum	cochlea
haemoglobin	liver	semicircular canals	stomach	thyroxine

Body processes	Substance or structure responsible
regulation of water levels in the blood	
conscious thought	
transport of oxygen	
urea formation	
control of metabolic rate	
detection of body position	
reduction of blood flow to the gut	

(Total for Question 1 = 7 marks)

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1. Question 1 · part whole · 7 marks · Working and answer

Working and answer

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2. Question 6 · part a.ii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.ii only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The effects of alcohol

Humans have been drinking alcohol for thousands of years. Drinking too much alcohol over many years can cause diabetes, malnutrition and diseases of the central nervous system and the liver. One immediate side effect of drinking too much alcohol is increased urination.

Beer is about 95% water and only 5% alcohol. The liver converts that 5% of alcohol into the same volume of water and some carbon dioxide. So if a person drinks 200 cm^3 of beer (one unit), the end result is 200 cm^3 of water. However, this person will not just urinate 200 cm^3 of urine, in fact the person urinates a total of 320 cm^3 of urine. So in general each unit of alcohol makes the person urinate an extra 120 cm^3 of urine more than the normal urine output.

A person with a mass of 60 kg produces 60 cm^3 of urine an hour and a person with a mass of 80 kg produces 80 cm^3 an hour and so on. Therefore if a 60 kg person drinks 200 cm^3 of beer in an hour they will produce 60 cm^3 of urine plus 320 cm^3, making a total of 380 cm^3.

Alcohol interferes with the mechanism that regulates water levels in our body because alcohol affects a small gland called the pituitary gland. One hormone released by the pituitary gland is anti-diuretic hormone (ADH). Alcohol reduces how much ADH is released by the pituitary. Each unit of alcohol that a person drinks forces the kidneys to produce an extra 120 cm^3 of urine more than the normal output.

Logic suggests that drinking lots of water would replace the 120 cm^3 of urine produced, but it is not that simple. The body only retains half the water that a person drinks. The other half leaves the body in urine, so the person will become dehydrated even if they drink lots of water.

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2. Question 6 · part a.ii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.ii only.

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(a) (i) Where in the body is the pituitary gland (line 16)?

(1)

- ☒ **A** brain
- ☒ **B** kidney
- ☒ **C** liver
- ☒ **D** pancreas

(ii) Which system includes the pituitary gland (line 16)?

(1)

- ☒ **A** digestive
- ☒ **B** endocrine
- ☒ **C** nervous
- ☒ **D** skeletal

(b) A person weighs 75 kg and drinks 400 cm^3 of beer in an hour (lines 11–14).

(i) Calculate the volume of urine produced by the person in this hour.

(3)

volume of urine = ... cm^3

(ii) Calculate the percentage increase in the volume of urine produced by this person in this hour.

(3)

percentage increase = ...%

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2. Question 6 · part a.ii · 1 mark · Working and answer

Working and answer

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3. Question 4 · part b.i · 1 mark

June 2021 Paper 2

Further practice · Attempt part b.i only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Obesity and diabetes have been linked to a 20% increase in the number of leg and foot amputations.

Data from Diabetes UK shows that there were 26 378 lower limb amputations related to diabetes from the period 2014 to 2017. This is an increase from the period 2010 to 2013 when there were 22 092 amputations related to diabetes.

Minor lower limb amputations are below the ankle and major lower limb amputations are below the knee. Minor lower limb amputations increased by 26.5% and major lower limb amputations increased by 4.0%.

In 2019 in the United Kingdom 3.8 million people had been diagnosed with diabetes and 90% of these people had Type 2 diabetes.

(a) (i) Describe the causes of obesity.

(3)

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(ii) Give two health risks, other than diabetes, associated with obesity.

(2)

1 .....

2 .....

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3. Question 4 · part b.i · 1 mark

June 2021 Paper 2

Further practice · Attempt part b.i only.

(b) Diabetes is caused as a result of insulin not being secreted or not working correctly.

(i) Which organ secretes insulin?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

(ii) Insulin stimulates the conversion of glucose into glycogen.

In which organ does this occur?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

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(c) (i) Calculate the percentage increase in the number of lower limb amputations between the period 2010 to 2013 and the period 2014 to 2017.

(3)

percentage increase = ...%

(ii) Calculate the number of people in 2019 who had been diagnosed with Type 2 diabetes in the United Kingdom.

(1)

number of people = ... million

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3. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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4. Question 3 · part a.iii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part a.iii only.

3 (a) In England, 3 319 266 people have diabetes.

(i) Give this value to 3 significant figures.

(1)

people with diabetes = ...

(ii) In the rest of the United Kingdom, 600 239 people have diabetes.

Give this value in standard form.

(2)

people with diabetes = ...

(iii) Diabetes can be caused by the inability of an organ to produce insulin.

Name the organ that produces insulin.

(1)

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4. Question 3 · part a.iii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part a.iii only.

(b) The body mass index (BMI) is a measure that uses height and body mass to work out if the body mass is healthy.

The table shows the BMI values and the categories into which a person can be placed.

BMI	BMI category
less than 18.5	underweight
18.5 to 24.9	healthy weight
25.0 to 29.9	overweight
30.0 to 39.9	obese
40.0 and over	morbidly obese

(i) Calculate the BMI of a person with a body mass of 120 kg and a height of 1.95 m.

Use the equation

$$\text{BMI} = \frac{\text{body mass}}{\text{height}^2}$$

(2)

BMI = ...

(ii) Determine the BMI category that this person belongs to.

(1)

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4. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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5. Question 3 · part f · 2 marks

June 2024 Paper 2

Further practice · Attempt part f only.

3 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

People make lifestyle choices that can sometimes have a negative effect on their health. People who drink too much alcohol are at risk of brain and liver damage, certain types of cancer and mental illness. Liver damage can affect the digestion of fat. Alcohol can also affect the eyes, increasing the risk of developing cataracts that cause the lenses in the eyes to become cloudy. Alcohol is also a depressant. It increases the activity of a chemical in parts of the brain that control balance, speech, motor skills and memory. In addition, alcohol suppresses an area of the brain called the amygdala. The amygdala is responsible for the fight or flight function which prepares the body for action, initiating the release of stress hormones. These hormones increase glucose usage by the brain and other parts of the body.

When a person stops drinking alcohol after years of drinking a lot of alcohol, the amygdala in the brain becomes fully active again, which is the cause of all forms of alcohol withdrawal. Over time, if a person regularly drinks heavily, the central nervous system gets used to the suppressing effect of the alcohol, which means the brain and body are affected if the alcohol level suddenly drops. An individual can go straight into fight or flight mode as the alcohol leaves the system and levels of stress hormones begin to return to normal. Symptoms of withdrawal can be as severe as seizures and hallucinations or less severe symptoms such as loss of appetite and insomnia (an inability to sleep).

(a) Drinking alcohol and diet are two lifestyle choices.

Describe how a lifestyle choice, other than drinking alcohol and diet, can have a harmful effect on health (Lines 1 and 2).

(2)

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5. Question 3 · part f · 2 marks

June 2024 Paper 2

Further practice · Attempt part f only.

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(b) Explain how drinking alcohol affects the digestion of fat (Lines 3 to 4).

(2)

(c) State how cataracts affect vision (Lines 4 to 5).

(1)

(d) Name a part of the brain, other than the amygdala, that is affected by alcohol (Lines 5 to 7).

(1)

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(e) Explain how alcohol acts as a depressant on the nervous system (Line 5).

(3)

5. Question 3 · part f · 2 marks

June 2024 Paper 2

Further practice · Attempt part f only.

(f) After a person stops drinking alcohol, the amygdala becomes fully active again.

The release of stress hormones, such as adrenaline, returns to normal.

Give two effects of adrenaline on the body.

(2)

1 .....

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2 .....

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(Total for Question 3 = 11 marks)

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Working and answer

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6. Question 7 · part c · 1 mark

June 2025 Paper 2R

Further practice · Attempt part c only.

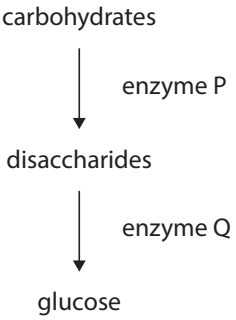
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7 (a) The diagram shows how carbohydrates are broken down to glucose by two different enzymes, enzyme P and enzyme Q.

These enzymes are found in different regions of the digestive system.



Name enzyme P.

(1)

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(b) People with diabetes can use glucose testing strips to test their urine.

Describe how enzymes are used in glucose testing strips.

(2)

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(c) People with type 1 diabetes are unable to produce hormones to control their blood glucose levels.

Name the organ responsible for producing hormones to control levels of glucose in the blood.

(1)

.....

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6. Question 7 · part c · 1 mark

June 2025 Paper 2R

Further practice · Attempt part c only.

(d) Insulin is a hormone used to treat people with diabetes.

Until recently insulin was obtained from animals for human use.

Suggest why this type of insulin can cause health complications in people with diabetes.

(2)

(e) Describe how yeast cells can be modified to produce the human form of insulin.

(2)

(Total for Question 7 = 8 marks)

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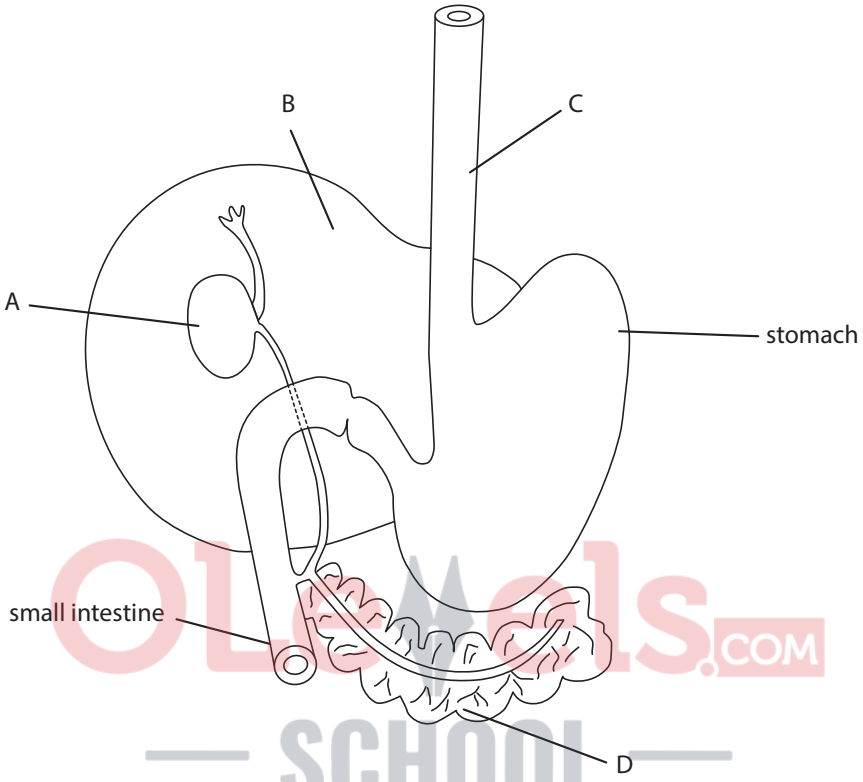
Working and answer

7. Question 2 · part b.ii · 2 marks

June 2025 Paper 2

Further practice · Attempt part b.ii only.

2 The diagram shows structures involved in digestion.



(a) (i) Identify the structures labelled A, B, C and D.

(4)

A .....

B .....

C .....

D .....

(ii) Name the fluid stored in structure A.

(1)

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7. Question 2 · part b.ii · 2 marks

June 2025 Paper 2

Further practice · Attempt part b.ii only.

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(iii) Explain the function of the fluid stored in structure A in digestion.

(3)

(b) (i) Describe the role of structure D in digestion.

(4)

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(ii) Name two substances secreted by structure D that are not involved in digestion.

(2)

1 .....

2 .....

(Total for Question 2 = 14 marks)

7. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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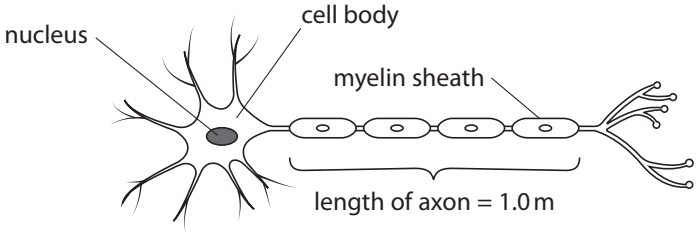
8. Question 5 · part b.i · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.i only.

5 The diagram shows a motor neurone.

not to scale



(a) (i) The actual diameter of the nucleus of the motor neurone is 12 µm.

A student observes the motor neurone using a ×50 microscope lens.

Calculate the diameter of the image seen after magnification.

Give your answer in millimetres.

(2)

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diameter = ... mm

(ii) Neurones transmit electrical impulses from one part of the body to another.

Suggest one function of the myelin sheath surrounding the axon.

(1)

.....

.....

(b) Nerve impulses travel at an average speed of 1.2×10^8 µm per second.

Hormones travel at an average speed of 1.5×10^6 µm per second.

(i) Explain this difference in speed.

(2)

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8. Question 5 · part b.i · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.i only.

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(ii) Calculate the time taken for a nerve impulse to travel along the axon of the motor neurone.

(3)

time taken = ... s

(Total for Question 5 = 8 marks)

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Working and answer

9. Question 6 · part a · 3 marks

January 2022 Paper 2

Further practice · Attempt part a only.

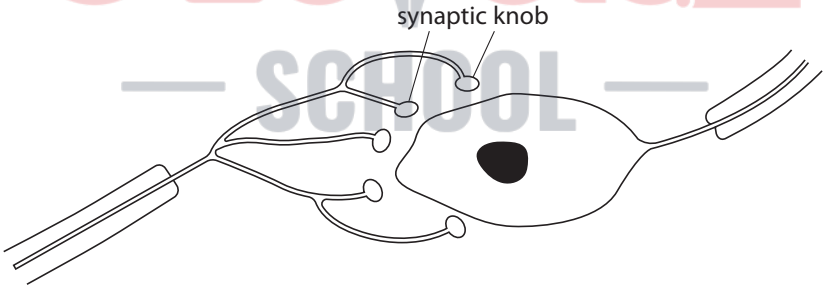
6 (a) Co-ordination of the activities of the body involves the nervous system and the hormonal system.

Complete the table to compare the two systems.

(3)

	Nervous system	Hormonal system
Route of transmission		
Type of signal		
Relative speed		

(b) The diagram shows a synapse.



(i) Draw an arrow on the diagram to show the direction an impulse travels.

(1)

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9. Question 6 · part a · 3 marks · Working and answer

Working and answer

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